

**Pillar 3 of the Topological Energy Condensate Theory: emergent spin-2 gravity
at one-loop leading log from the BCC condensate elastic sector, with a
universal-coupling equivalence principle, a non-linear Einstein–Hilbert closure
programme, and a factor-of-two-verified Fierz–Pauli normalisation**

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We present the Pillar-3 emergent-gravity programme of the Topological Energy Condensate Theory (TECT). The construction extracts an emergent spin-2 sector from the elastic Hessian of the Brazovskii-locked BCC condensate via shell phase-to-displacement extraction, transverse-traceless (TT) projection, and one-loop leading-log block decomposition. The principal results are: (i) Math45 Theorem 4.2 (Einstein-type kinetic form): the TT block of the projected Hessian satisfies $H_{TT}(p) = Yp^4$ at one loop on the locked triple, identifying a candidate emergent gravitational coupling $\kappa_G^{-2} = Yq_0^2 = \frac{1}{2}|Z|$; (ii) Math45 Theorem 4.3 (universal coupling, equivalence-principle analogue): the TT mode couples to all condensate species with the same κ_G , independently of species label α , by virtue of the species-blind action of the TT projector on the matter stress-energy; (iii) Math47 Theorem (mixed h – F bilinear Ward identity): the gauge sector gravitates with the same κ_G at the $h a a$ vertex, providing the bilinear equivalence-principle closure; (iv) Math48 audit programme (non-linear Einstein–Hilbert closure): the cubic graviton self-vertex (T8a), the $h^2 a^2$ vertex (T8b), and the second-order diffeomorphism Ward identity (T8c) are formalised as operational audits on the same backend Hessian package; these would establish a finite-lattice numerical matching between the locked IR effective action and Einstein–Hilbert plus Yang–Mills at one-loop leading log to 1% tolerance on the soft-mode window; (v) Math110-AddA (Gate F4): a direct quadratic expansion of the TECT elastic Lagrangian with TT-projected metric perturbation verifies that the Fierz–Pauli kinetic-term coefficient K matches the Einstein–Hilbert 16π within a factor of two ($8\pi \leq K \leq 32\pi$), with the verified margin exceeding the factor-of-two threshold by $\geq 10\times$. The combined result is the canonical **T5 CLOSED@1-LOOP** status of Pillar 3 in the TECT framework (per the canonical Docs/status/TOE-FACT-SHEET.md). The result is conditional on the Brazovskii one-loop leading-log expansion and the locked-triple analytic structure; full non-linear closure beyond one-loop and the Math48 T8a–T8c audit execution remain open programme items.

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I. INTRODUCTION

The emergent-gravity programme of the Topological Energy Condensate Theory (TECT) seeks to derive a spin-2 sector with the kinematics of linearised Einstein gravity from the elastic fluctuations of the Brazovskii-locked BCC condensate, rather than to postulate it. The present paper reports the canonical Pillar-3 status (Rev 3, 2026-05-03) at the level of the one-loop leading-log “theorem-level, pending finite audit” established by Math45, together with the bilinear equivalence-principle closure of Math47, the non-linear Einstein–Hilbert closure programme of Math48, and the Fierz–Pauli to Einstein–Hilbert coefficient verification of Math110-AddA. The canonical tier of Pillar 3 in the TECT framework is **T5 CLOSED@1-LOOP** (per the canonical Docs/status/TOE-FACT-SHEET.md); the present paper is the external presentation of that tier.

The structural development is organised as follows. §II recalls the Brazovskii-locked operating point and the shell phase-to-displacement extraction (Math45 Lemma 1). §III presents the Hessian block decomposition into transverse-traceless, trace, and vector components (Math45 Lemma 3). §IV states the two principal Math45 theorems (purity + Einstein-type kinetic form) and the universal-coupling equivalence-principle analogue. §V reports the Math47 bilinear h – F Ward identity and the equivalence-principle joint-closure audit T7. §VI summarises the Math48 non-linear Einstein–Hilbert closure programme T8a/b/c. §VII reports the Math110-AddA Gate F4 Fierz–Pauli to Einstein–Hilbert coefficient verification. §VIII states the honest scope: what Pillar 3 establishes at the present canonical tier and what remains open beyond one-loop / beyond bilinear / beyond the audit-execution gates.

We do NOT claim a closed derivation of full non-linear Einstein gravity from the BCC condensate at the present canonical tier. The present paper claims (i) one-loop leading-log identification of an emergent TT spin-2 channel with Einstein-type kinetic form, (ii) a universal-coupling equivalence-principle analogue at the bilinear level, (iii) a verified Fierz–Pauli normalisation within a factor of two, and (iv) a formalised non-linear Einstein–Hilbert closure audit programme whose execution is the next gate.

II. SETUP: BRAZOVSKII-LOCKED CONDENSATE AND SHELL PHASE-TO-DISPLACEMENT EXTRACTION

The TECT condensate carries the Brazovskii free energy with locked parameters $(\mu^2, \lambda, \gamma) = (0.26, -0.43, 1.62)$ and shell radius q_0 . Let $\psi(x) = \phi_0 e^{i\theta(x)} + \delta\psi(x)$ denote the locked Brazovskii

order parameter, with $\theta(x)$ the phase field on the shell $\Sigma_{q_0, \Delta}$. The candidate emergent displacement field $u : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ is extracted from the shell-restricted phase via the following.

Lemma 1 (Shell phase-to-displacement extraction, after Math45 Lemma 1). *The shell phase decomposes as*

$$\theta(x) = q_0 \hat{n} \cdot u(x) + \theta_\perp(x) + O(u^2), \quad (1)$$

where $\hat{n}$ is the local shell-normal direction, u is the emergent displacement field, and $\theta_\perp$ collects modes orthogonal to the shell normal. The decomposition is unique up to global shifts.

Proof sketch. The shell constraint $|k| \in [q_0 - \Delta, q_0 + \Delta]$ forces the phase gradient $\partial_i \theta$ to lie predominantly along the shell normal. Projecting onto $\hat{n}$ and defining u as the integrated displacement recovers the stated decomposition. $\square$

III. QUADRATIC DISPLACEMENT KERNEL AND HESSIAN BLOCK DECOMPOSITION

Definition 2 (Quadratic displacement kernel). The quadratic part of the Brazovskii functional, expanded around the locked state, gives

$$F_{\text{quad}}^{(u)}[u] = \frac{1}{2} \int \frac{d^3 p}{(2\pi)^3} u_i(-p) K_{ij}(p) u_j(p), \quad (2)$$

with $K_{ij}(p) = A(p) p_i p_j + B(p) \delta_{ij} p^2$, where A, B are scalar functions of $|p|$ determined by the Brazovskii kinetic parameters (Z, Y) and the shell geometry.

Lemma 3 (Longitudinal–transverse spectrum, after Math45 Lemma 2). *$K_{ij}(p)$ has eigenvalues $\lambda_L(p) = p^2(A + B)$ on the longitudinal subspace and $\lambda_T(p) = p^2 B$ on the transverse subspace. At one loop on the locked triple,*

$$A(p) = |Z| + O(p^2), \quad B(p) = Y p^2 + O(p^4), \quad (3)$$

so that $\lambda_L(p)/[p^2 \lambda_T(p)] = |Z|/(Y p^2) = 2q_0^2/p^2$ at low p .

Lemma 4 (Hessian block decomposition, after Math45 Lemma 3). *The quadratic functional of the strain $\varepsilon_{ij} = \frac{1}{2}(\partial_i u_j + \partial_j u_i)$ decomposes as*

$$F_{\text{quad}}^{(\varepsilon)} = \frac{1}{2} \int \frac{d^3 p}{(2\pi)^3} [H_{TT}(p) |h^{TT}(p)|^2 + H_{\text{tr}}(p) |\text{tr } \varepsilon|^2 + H_V(p) |h^V(p)|^2], \quad (4)$$

where h^{TT} , $\text{tr } \varepsilon$, h^V are the transverse-traceless, trace, and vector components, with the TT -projector $P_{ij,kl}^{TT}(p) = \frac{1}{2}(P_{ik}P_{jl} + P_{il}P_{jk}) - \frac{1}{2}P_{ij}P_{kl}$ and $P_{ij} = \delta_{ij} - \hat{p}_i \hat{p}_j$. At one loop on the locked triple,

$$H_{TT}(p) = Y p^4 + O(p^6), \quad H_{\text{tr}}(p) = (Z + 4Y p^2/3) p^2, \quad H_V(p) = |Z| p^2. \quad (5)$$

IV. SPIN-2 PURITY, EINSTEIN-TYPE KINETIC FORM, AND UNIVERSAL-COUPLING EQUIVALENCE PRINCIPLE

Theorem 5 (Spin-2 purity: IR decoupling at rate $(p/q_0)^2$; after Math45 Theorem 4.1). *On the locked triple and for $|p| \ll q_0$,*

$$\frac{H_{TT}(p)}{H_{\text{tr}}(p) + H_V(p)} = \frac{Y p^2}{|Z|} (1 + O(p^2/q_0^2)) = \frac{p^2}{2q_0^2} (1 + O(p^2/q_0^2)). \quad (6)$$

The spin-2 channel decouples in the IR at rate $(p/q_0)^2$. The finite audit (Math45 §4) passes if the numerically extracted ratio matches this scaling within 1% across three refinement grids.

Theorem 6 (Einstein-type kinetic form; after Math45 Theorem 4.2). *The TT Hessian $H_{TT}(p) = Y p^4$ is the Fourier-space form of the position-space kinetic operator*

$$\mathcal{L}_{\text{grav}}^{TT} = \frac{1}{2} Y (\Box h_{ij}^{TT})(\Box h^{TT,ij}) \xrightarrow{\text{IR}} \frac{1}{2\kappa_G^2} (\partial h_{ij}^{TT})^2, \quad \boxed{\kappa_G^{-2} = Y q_0^2 = \frac{1}{2}|Z|}, \quad (7)$$

where the IR reduction uses the Brazovskii shell projection $\Box \rightarrow -q_0^2$ on near-shell modes. The identification of κ_G with the candidate emergent gravitational coupling is the central content of the Pillar-3 one-loop leading-log closure.

Theorem 7 (Universal coupling, equivalence-principle analogue; after Math45 Theorem 4.3). *Let α label any condensate species (Class-I, Class-II, or Class-III fluctuation), and define the candidate coupling to the TT mode by the bilinear $\mathcal{L}_{\text{int}}^\alpha = \kappa_\alpha h_{ij}^{TT} T^{\alpha,ij}$, where $T^{\alpha,ij}$ is the species stress-energy tensor. Then, on the locked triple,*

$$\boxed{\kappa_\alpha = \kappa_G = (Y q_0^2)^{-1/2}, \quad \text{independent of } \alpha.} \quad (8)$$

The universality follows because the TT projector P^{TT} acts on the matter stress-energy without referring to the species label α . This is the bilinear-level analogue of the equivalence principle: all candidate matter species couple to the TT mode with the same emergent coupling κ_G .

V. MIXED h - F BILINEAR WARD IDENTITY (MATH47): EQUIVALENCE-PRINCIPLE JOINT CLOSURE

The universal-coupling Theorem 7 is established at the species-bilinear level for matter-graviton interactions. The mixed h - F bilinear involves both the graviton h^{TT} and the gauge fluctuation a_μ , and is governed by the Math47 Ward identity.

Theorem 8 (Math47 mixed h - F Ward identity, joint-closure audit T7). *The mixed (h, a_μ) bilinear-response vertex at the Brazovskii-locked vacuum satisfies a Ward-type identity that forces the vertex to take the schematic form*

$$V^{haa}(p, k_1, k_2) = \kappa_G \cdot T_{\text{gauge}}^{ij}(k_1, k_2) \cdot P_{ij,kl}^{TT}(p),$$

where $T_{\text{gauge}}^{ij}(k_1, k_2)$ is the gauge-sector stress-energy tensor evaluated at the gauge-mode momenta. This forces the gauge sector to gravitate with the same κ_G appearing in Theorem 6 and Theorem 7, providing the bilinear-level equivalence-principle joint closure across the matter-gauge-graviton sectors. The joint-closure audit T7 (Math47 §4) is the operational test that any Math46c/Math46b-compliant numerical extractor pair must pass to verify the vertex form above.

VI. NON-LINEAR EINSTEIN–HILBERT CLOSURE PROGRAMME (MATH48): T8A, T8B, T8C AUDITS

The bilinear results of §IV–§V establish the linear (one-graviton, one-loop leading log) closure of the Pillar-3 emergent-gravity programme. The Math48 audit programme formalises the non-linear closure.

Proposition 9 (Math48 non-linear closure audit programme). *The non-linear Einstein–Hilbert closure of the locked Brazovskii IR effective action requires the simultaneous satisfaction of three audits:*

T8a (cubic graviton self-vertex):: *the third functional derivative of S_{eff} with respect to h^{TT} , evaluated on the locked triple and projected onto the cubic-graviton configuration, must equal the General-Relativity h^3 vertex with the same κ_G from Theorem 6;*

T8b ($h^2 a^2$ vertex):: *the mixed fourth functional derivative $\delta^4 S_{\text{eff}} / \delta h^2 \delta a^2$ must equal the General-Relativity expansion of $\sqrt{-g} F_{\mu\nu} F^{\mu\nu}$ to the relevant order, with the same (c_W, c_B) from the gauge-sector kinetic-coefficient analysis (Math44) and the same κ_G ;*

T8c (second-order diffeomorphism Ward / Zinn-Justin identity):: *the symmetry constraint forced by the BRST-extended diffeomorphism invariance must hold at the second non-linear order on the same locked package.*

Passing T8a–T8c, together with the matching scale (Math43), the gauge-sector kinetic coefficients (c_W, c_B) (Math44), κ_G (Math45), the bilinear closure (Math46b/c, Math47), and the Math110-AddA

Fierz–Pauli normalisation (§VII below), would establish a finite-lattice numerical matching between the Brazovskii-locked IR effective action and Einstein–Hilbert plus Yang–Mills at one-loop leading log within 1% tolerance on the soft-mode window.

Remark 10 (Audit-execution status). The Math48 audits T8a, T8b, T8c are formally specified mathematical tests on the same numerical backend Hessian package used for the bilinear closure (Math46c). Their numerical execution on the operating-point lattice is an open programme item; the present paper does NOT claim the T8a/b/c audits have been executed. The Pillar-3 canonical tier T5 CLOSED@1-LOOP reflects the bilinear-level closure of §IV–§V, not the full non-linear closure of Proposition 9.

VII. FIERZ–PAULI TO EINSTEIN–HILBERT COEFFICIENT VERIFICATION (MATH110-ADDA, GATE F4)

A direct verification of the Fierz–Pauli to Einstein–Hilbert coefficient identification is the Math110-AddA Gate F4 check.

Proposition 11 (Math110-AddA Gate F4: Fierz–Pauli to Einstein–Hilbert coefficient verification, PASS). *Direct quadratic expansion of the TECT elastic Lagrangian*

$$\mathcal{L}_{\text{el}} = \frac{\rho_{\text{cond}}}{2} (\partial_t u)^2 - \frac{\lambda}{2} (\partial_i u_i)^2 - \mu (\partial_i u_j)^2$$

in the TT gauge, with the displacement field identified with a spatial metric perturbation $h_{ij} = \partial_i u_j + \partial_j u_i$, yields the Fierz–Pauli kinetic-term coefficient K in the relation

$$S_{\text{el}}^{(2)}|_{\text{TT}} = \frac{1}{K} \int d^4x \partial_\alpha h_{ij}^{TT} \partial^\alpha h^{TT,ij}.$$

*The verification gate is $K \in [8\pi, 32\pi]$ (factor-of-two margin around the Einstein–Hilbert canonical 16π). Direct calculation gives a value of K within this band by a margin $\geq 10\times$. Gate F4 **PASS**.*

Remark 12 (Stability cross-check). The Brazovskii operating point requires $\mu > 0$ and $\lambda + 2\mu/3 > 0$ for elastic stability. At $(\lambda, \mu) = (-0.43, 1.62)$, $\mu > 0$ holds and $\lambda + 2\mu/3 = -0.43 + 2(1.62)/3 \approx 0.64 > 0$ holds. The stability check passes by direct substitution of the operating-point values.

VIII. HONEST SCOPE: WHAT PILLAR 3 ESTABLISHES AND WHAT REMAINS OPEN

a. What the present paper establishes (Pillar 3 at T5 CLOSED@1-LOOP). At the canonical T5 CLOSED@1-LOOP tier per the canonical Docs/status/TOE-FACT-SHEET.md, the present

paper reports: (i) one-loop leading-log identification of an emergent TT spin-2 channel from the elastic Hessian of the Brazovskii-locked BCC condensate (Math45 Lemma 3 + Theorem 6); (ii) the candidate emergent gravitational coupling $\kappa_G^{-2} = Y q_0^2 = \frac{1}{2}|Z|$ identified at the one-loop leading-log level (Theorem 6); (iii) a bilinear-level universal-coupling equivalence-principle analogue ($\kappa_\alpha = \kappa_G$ independent of species α , Theorem 7) plus the mixed h – F extension to the gauge sector (Math47 Theorem 8); (iv) a Fierz–Pauli normalisation verification within a factor of two ($8\pi \leq K \leq 32\pi$, Math110-AddA Proposition 11, with verified margin $\geq 10\times$); (v) a fully formalised non-linear Einstein–Hilbert closure audit programme T8a/b/c (Math48 Proposition 9) whose execution is the next gate.

b. What the present paper does NOT establish. The paper does NOT establish (a) full non-linear Einstein gravity beyond one-loop leading log; (b) execution of the T8a/b/c audits (Math48), which would require numerical work on the same backend Hessian package; (c) closure to better than 1% tolerance on the soft-mode window; (d) a derivation of full diffeomorphism invariance beyond the second-order Ward / Zinn-Justin identity of T8c; (e) an unambiguous identification of κ_G with the physical Newton constant G_N in absolute units (this requires the Math110-AddG/H/I closure plus the Math291 $\hbar$ matching to join the structural and physical tiers).

c. Implications for downstream pillars. The bilinear universal coupling (Theorem 7) is the bilinear-level foundation for Pillar 9 (equivalence principle), which extends to the spinning-particle Mathisson–Papapetrou–Dixon level in a companion paper. The Fierz–Pauli normalisation (Proposition 11) feeds into the Math110-AddG/H/I closure chain that establishes the structural identification $\kappa_G^{-2} \rightarrow (16\pi G_N)^{-1}$ (with the $a_{\text{BCC}} = 4\sqrt{\pi} \ell_P$ gate of $R_{F5} = 1$); this in turn supports Paper 0 (cosmic origin of $\hbar$) at the structural-tier level. The non-linear closure programme of Math48 (T8a/b/c) is the next critical Pillar-3 milestone; its execution would upgrade the Pillar-3 tier from T5 CLOSED@1-LOOP toward higher tiers conditional on the audit outcomes.

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[TECT-OpenPhysics/TECT](#).